Mingyue Xu

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EDUCATION

Purdue University West Lafayette, IN
Ph.D. Student in Computer Science Aug 2023 – Present
Advisor: Steve Hanneke

Columbia University New York, NY
M.S. in Data Science Sep 2021 – Dec 2022
Advisor: Daniel Hsu | GPA: 3.73/4.00

Shanghai Jiao Tong University Shanghai, China
B.S. in Mathematics Sep 2017 – Jun 2021
GPA: 3.62/4.00

RESEARCH INTERESTS

Statistical learning theory, algorithmic statistics, deep learning theory, and foundations of artificial intelligence.

PUBLICATIONS & PREPRINTS

α-β indicates alphabetical author order; * indicates equal contribution.

[1] S. Hanneke, H. Wang, and **M. Xu** (α-β). "Towards a theory of inference-time alignment with unknown rewards." *Preprint*.

[2] D. Hsu and **M. Xu** (α-β). "Attention-based representations for multi-task computation." *Preprint*.

[3] S. Hanneke, H. Wang, and **M. Xu** (α-β). "Optimal Learning Under Tsybakov Noise." *Preprint*.

[4] S. Hanneke and **M. Xu** (α-β). "When More Data Doesn't Help: Limits of Adaptation in Multitask Learning." *The Forty-third International Conference on Machine Learning (ICML 2026)*.

[5] **M. Xu**, G. Vardi, and I. Safran. "To Grok Grokking: Provable Grokking in Ridge Regression." *The Forty-third International Conference on Machine Learning (ICML 2026)*. **Oral presentation; Outstanding Paper Honorable Mention**.

[6] S. Hanneke and **M. Xu** (α-β). "Universal Rates of ERM for Agnostic Learning." *The Thirty-eighth Annual Conference on Learning Theory (COLT 2025)*.

[7] G. Yuan*, **M. Xu***, S. Kpotufe, and D. Hsu. "Efficient Estimation of the Central Mean Subspace via Smoothed Gradient Outer Products." *SIAM Journal on Mathematics of Data Science (SIMODS 2025)*.

[8] S. Hanneke and **M. Xu** (α-β). "Universal Rates of Empirical Risk Minimization." *The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024)*.

PUBLICATIONS & PREPRINTS (CONTINUED)

[9] J. Tu, W. Liu, X. Mao, and **M. Xu**. "Distributed Semi-Supervised Sparse Statistical Inference." *IEEE Transactions on Information Theory (TIT 2023)*.

WORK EXPERIENCE

Machine Learning Applied Research Scientist Intern — Nov 2020 – Jun 2021
Qutronix Co., Ltd. — Shanghai, China

AWARDS

ICML 2026 Outstanding Paper Honorable Mention — Jul 2026
Purdue University

Excellent Bachelor Thesis Award — Jun 2021
School of Mathematical Sciences, Shanghai Jiao Tong University

Second Prize of Undergraduate Mathematical Contest in Modeling — Oct 2018
Regional Final, Shanghai, China

SERVICE

International Conference on Learning Representations (ICLR) 2026, 2027
Reviewer

International Conference on Machine Learning (ICML) 2026
Reviewer
- **Recognized as a Gold Reviewer**

Neural Information Processing Systems (NeurIPS) 2026
Reviewer

TEACHING EXPERIENCE

CS 182: Foundations of Computer Science — Spring 2025, Spring 2026
Teaching Assistant — Purdue University

CS 251: Data Structures and Algorithms — Fall 2023, Spring 2024, Fall 2024, Fall 2026
Teaching Assistant — Purdue University

CS 381: Introduction to the Analysis of Algorithms — Fall 2025
Teaching Assistant — Purdue University